Sayan Deb Sarkar

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EDUCATION

Since 2024 PhD in Computer Vision, *Stanford University*, United States
Advised by Prof. Iro Armeni, Gradient Spaces Research Group (part of Stanford Vision and Learning Lab).

2022 - 2024 MSc in Computer Science, *ETH Zürich*, Switzerland
Advised by Prof. Marc Pollefeys, Computer Vision And Geometry Group. GPA: 5.48/6.0

2016 - 2020 B.Tech in Information Technology, *Manipal Institute of Technology*, India
Relevant Coursework: Data Structures, Operating Systems. GPA: 9.16/10.0 ≈ top 1%

EXPERIENCE

Summer 2026 Research Intern at **Waymo Perception**, *Mountain View*, USA
Developed a streaming recurrent temporal approach for VideoLMs using self-supervised future prediction objectives to improve fine-grained motion understanding.
Mentor: Chenyun Wu (Semantics Foundation) and Nikhil Parthasarathy (Google Deepmind)

Summer 2025 Research Scientist Intern at **Microsoft Spatial AI Lab**, *Zurich*, Switzerland
Designed a scalable and efficient tokenization method for Video LLMs using codec information to enable longer context and faster video understanding.
Mentor: Prof. Marc Pollefeys

Autumn 2023 Research Intern at **Qualcomm XR Labs**, *Amsterdam*, Netherlands
Optimized SLAM algorithms for real-time performance for extented reality applications & improved tracking in adversarial scenarios.
Mentor: Marco Manfredi

2022 - 2024 Research Student at **CVG, ETH Zürich**, *Zürich*, Switzerland
3D scene graph alignment in static and dynamic environments, leverage the graph matching to enable embodied agent tasks like map reuse, 3D localization and registration.
Supervisor: Dániel Béla Baráth, Ondrej Miksik & Prof. Iro Armeni

2021 - 2022 Computer Vision Research Engineer at **Mercedes-Benz R & D**, *Bangalore*, India
Developed deep learning models for driver monitoring and head position estimation in multi-purpose camera systems for the Maybach S-Class under the Interior Assist program.

2020 - 2021 Research Engineer at **IVC, TU Graz**, *Graz*, Austria
Joint 3D hand + object pose estimation in close interaction scenarios and indoor 3D scene understanding estimation using Monte Carlo Tree Search on noisy RGB-D scans.
Supervisor: Shreyas Hampali, Mahdi Rad & Prof. Vincent Lepetit

SELECTED PUBLICATIONS

[8] CoPE-VideoLM: Leveraging codec Primitives For Efficient Video Language Modeling, *in* **NeurIPS 2026**.
Sayan Deb Sarkar, Rémi Pautrat, Ondrej Miksik, Marc Pollefeys, Iro Armeni, Mahdi Rad*, and Mihai Dusmanu*
[Paper] [Project Page]

[7] GuideFlow3D: Optimization-Guided Rectified Flow For Appearance Transfer, *in* **NeurIPS 2025**.
Sayan Deb Sarkar, Sinisa Stekovic, Vincent Lepetit, and Iro Armeni
[Paper] [Project Page]
Featured: Voxel51.

[5] CrossOver: 3D Scene Cross-Modal Alignment, *in* **CVPR 2025** [Highlight, top 3%].
Sayan Deb Sarkar, Ondrej Miksik, Marc Pollefeys, Dániel Béla Baráth, and Iro Armeni
Featured: Open Robotics.
[Paper] [Project Page]

[4] SGAligner: 3D Scene Alignment with Scene Graphs, *in* **ICCV 2023**.
Sayan Deb Sarkar, Ondrej Miksik, Marc Pollefeys, Dániel Béla Baráth, and Iro Armeni
Featured: Computer Vision News, Learn OpenCV Blog.
[Paper] [Project Page]

[3] Keypoint Transformer: Solving Joint Identification in Challenging Hands and Object Interactions for Accurate 3D Pose Estimation, *in* **CVPR 2022** [Oral, top 4.1%].
Shreyas Hampali, **Sayan Deb Sarkar**, Mahdi Rad, and Vincent Lepetit
[Paper] [Project Page]

[2] Monte Carlo Scene Search For 3D Scene Understanding, *in* **CVPR 2021**.
Sinisa Stekovic*, Shreyas Hampali*, **Sayan Deb Sarkar**, Chetan Srinivasa Kumar, Friedrich Fraundorfer, and Vincent Lepetit
[Paper] [Project Page]

[1] General 3D Room Layout from a Single View by Render-And-Compare, *in* **ECCV 2020**.
Sinisa Stekovic, Shreyas Hampali, Mahdi Rad, **Sayan Deb Sarkar**, Friedrich Fraundorfer, and Vincent Lepetit
[Paper] [Project Page]

PATENTS

[B] Efficient Video Tokenization for Multi-modal Models
US patent, filed in 2025, by Microsoft.

[A] Learned Occlusion Modeling For Simultaneous Localization and Mapping
US patent, filed in 2024, by Qualcomm. [Patent Link]

INVITED TALKS

2026 Codec Primitives for Efficient Video Understanding. *Google Deepmind & Valeo AI.*

2026 Optimization-Guided Rectified Flow For Appearance Transfer. *Voxel51, Best of NeurIPS.*

2025 Scalable Cross-Modal 3D Scene Understanding. *Google XR Research & Imagine Labs, ENPC ParisTech.*

ACADEMIC SERVICES

Reviewing CVPR, ECCV, ICCV, NeurIPS, ICRA

Organization CV4AEC Workshop@CVPR ('23, '24, '26)

Recognition Outstanding Reviewer (CVPR '26)

TEACHING

Winter 2025 / 2026 Head TA, Computer Vision For The Built Environment [Course Website]

TECHNICAL SKILLS

Programming Python, C++, Java, JavaScript

Tools Pytorch, Tensorflow, Blender, OpenCV, mySQL, Node.js, Django, mongoDB

EXTRA CURRICULAR

2022 Co-founder, *CORD.ai*
Built and led a core team of 14 to establish a 350+ member community focused on democratizing AI, reducing barriers for young independent researchers, and fostering collaboration.

2020 Technical Head, *defeatCOVID*
Non-profit organisation, aimed at tracking the spread of COVID-19 using a mobile-based heat map interface.